Chronic HBV and HCV Basics and Managements

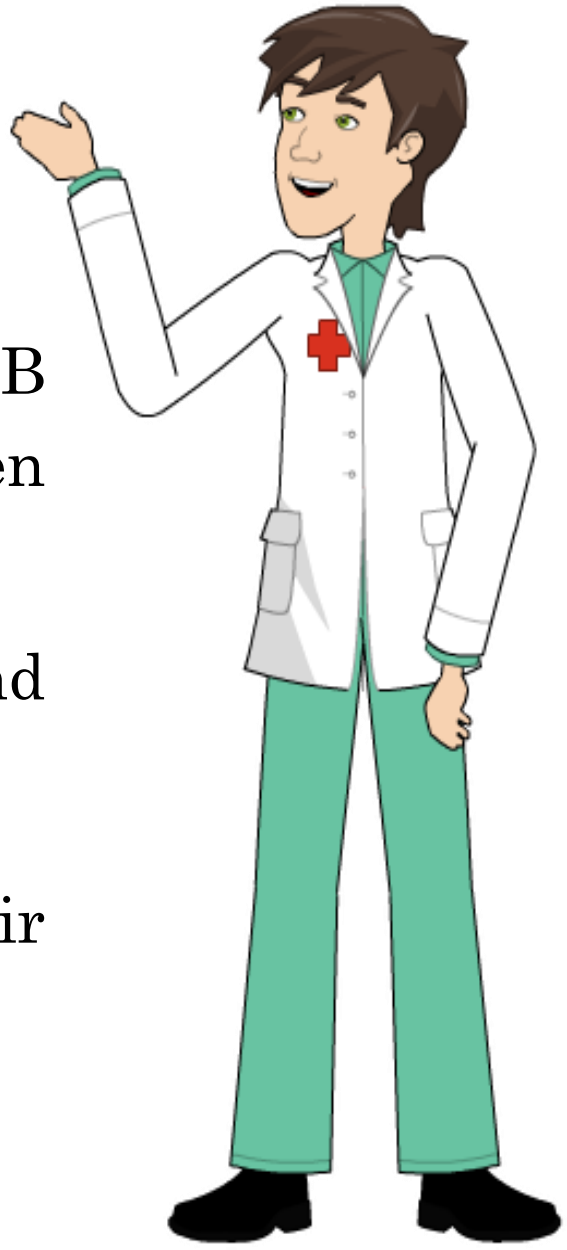
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Chronic HBV

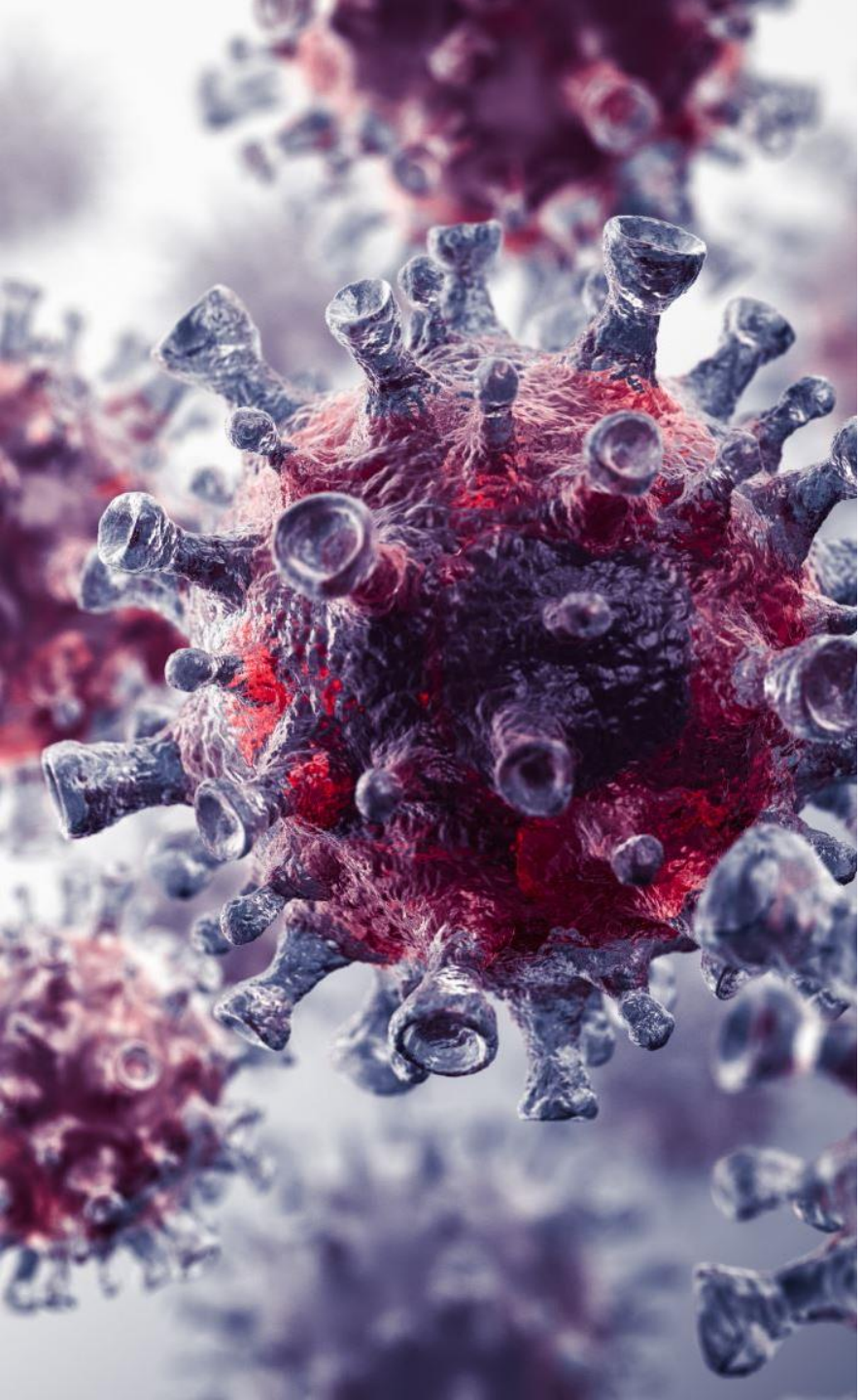
EPIDEMIOLOGY

- In 2015, WHO estimates that
  - 257 million people were living with chronic hepatitis B infection (defined as hepatitis B surface antigen positive).
  - 887 000 HBV related-deaths, mostly from cirrhosis and hepatocellular carcinoma (i.e. primary liver cancer).
- As of 2016, 27 million people (10.5%) were aware of their infection, while 4.5 million (16.7%) were on treatment.



EPIDEMIOLOGY

- In Egypt,
  - WHO estimated that 3.3 million persons are infected with hepatitis B.
  - While in 2015 survey. population HBV prevalence was 1.4%, with a HBV-HCV co-infection rate of 0.06%

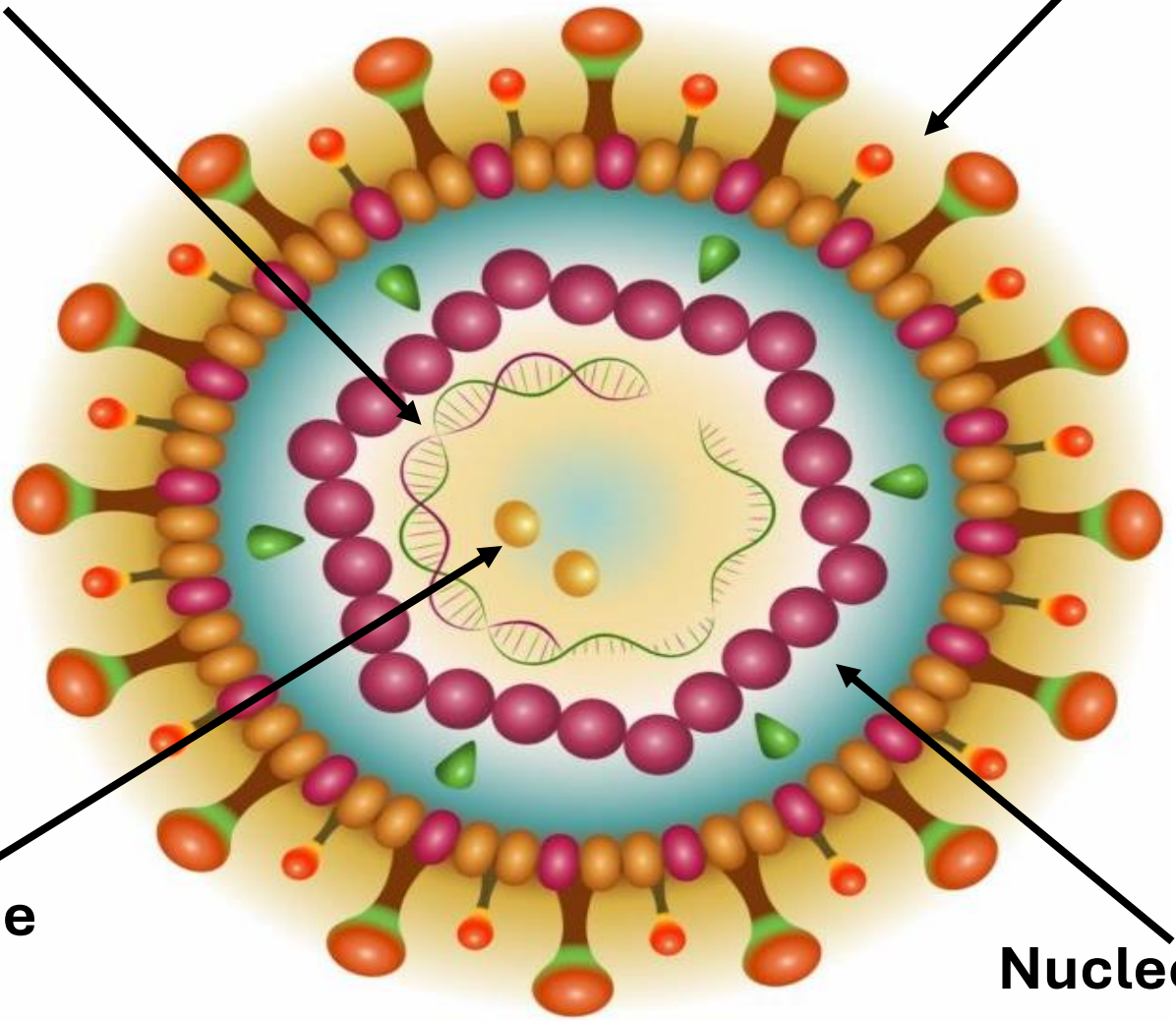


Virology

- It is a member of the Hepadnaviridae family, is a small DNA virus with unusual features similar to retroviruses. HBV replicates through an RNA intermediate and can integrate into the host genome.

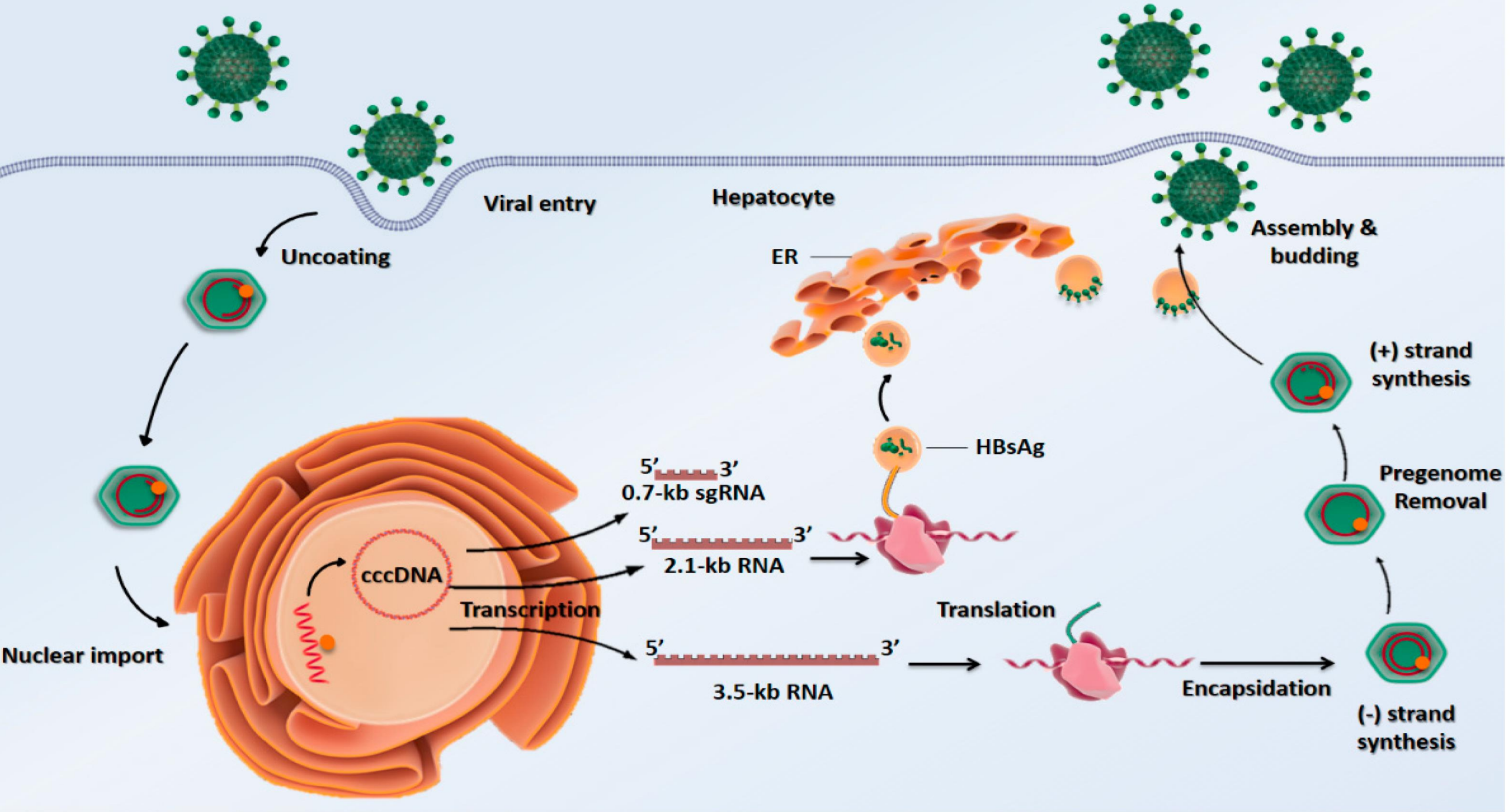
Partially doubled DNA

Lipid layer with HBsAg



RT/ DNA polymerase

Nucleocapsid with HBc Ag and Hbe Ag


Viral entry
Hepatocyte
Uncoating
ER
Assembly &
budding
(+) strand
synthesis
HBsAg
Pregenome
Removal
5′ 3′
0.7-kb sgRNA
5′ 3′
2.1-kb RNA
cccDNA
Transcription
Translation
Nuclear import
5′ 3′
3.5-kb RNA
Encapsidation
(-) strand
synthesis

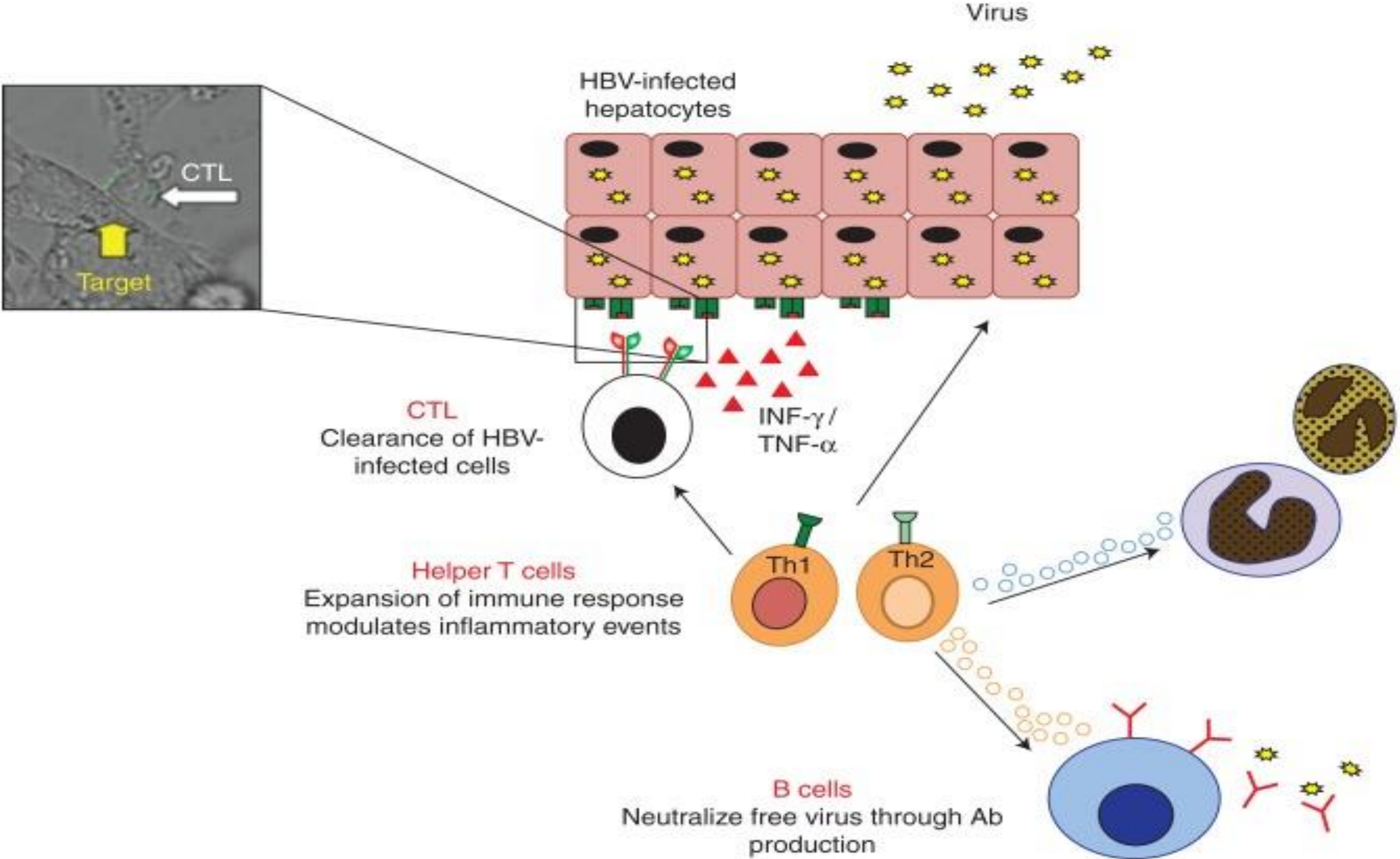
IMMUNE RESPONSE

- HBV stimulate both innate and adaptive immune system.
- In acute resolving infections, the response is efficient and timely.
- Viral clearance involves the induction of a robust adaptive T cell reaction inducing both:

    1. Cytolytic dependent and independent antiviral effect via the expression of antiviral cytokines.
    2. B cells producing neutralizing antibodies preventing the spread of the virus.

IMMUNE RESPONSE

- Hepatocyte turnover resulting from infected cell death leads to cccDNA dilution.
- When the acute infection becomes chronic, there is a progressive impairment in HBV specific T cell function.


Virus
HBV-infected hepatocytes
CTL
Target
CTL
Clearance of HBV-infected cells
INF-γ / TNF-α
Th1
Th2
Helper T cells
Expansion of immune response modulates inflammatory events
B cells
Neutralize free virus through Ab production

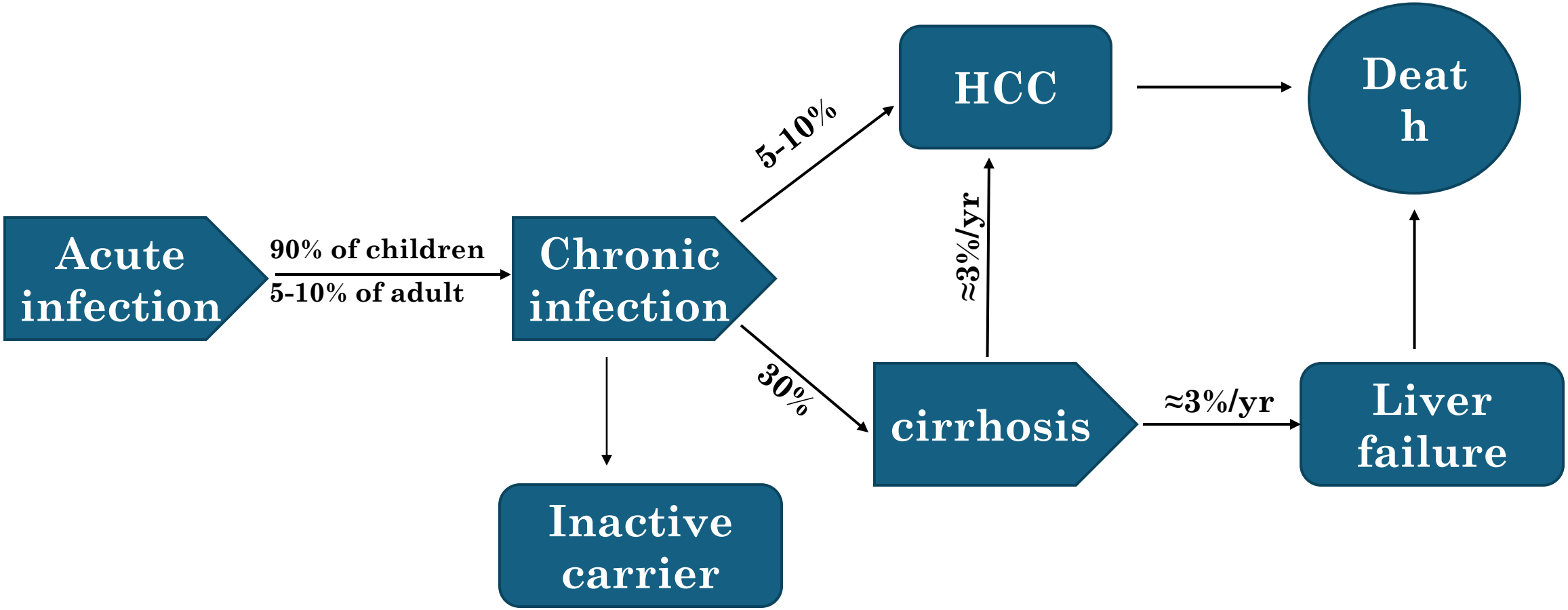
TRANSMISSION

Vertically	•Mother to child during childbirth. •Common in high prevalence countries (in the absence of infant vaccination). •HBeAg-seropositive mothers or with high viral loads Up to 85% of their infants to became chronically infected, while 32% of those born to HBeAg-seronegative mothers
Horizontally	
• **House hold**	•From an individual with HBV to unvaccinated household contacts (e.g. sharing toothbrushes, razors, other items that may lead to exchange of body fluids).
• **Sexually**	•Unprotected vaginal, anal or oral sex with a person who has HBV.

TRANSMISSION

Horizonatlly	
• **Percutaneously**	•Sharing/re-use of injecting equipment. •tattooing or body piercing including acupuncture, cupping and other practices involving blood or bodily fluids.
• **Medically**	•Medical procedures (e.g. dentistry, surgery, dialysis) pose a risk if appropriate infection control procedures are not adhered to. •Needlestick injury with risk of transmission about 30% (10 times HCV and 100 time HIV) •splashing of infected blood or body fluids.

NATURAL HISTORY



CLINICAL MANIFESTATION

- Acute hepatitis
- Fulminant hepatitis 0.5%-1%
- Chronic hepatitis
- Cirrhosis
- Hepatocellular carcinoma

CLINICAL MANIFESTATION

- Extrahepatic manifestations include
  - Vasculitis,
  - Skin manifestations (purpura),
  - Polyarteritis nodosa,
  - Arthralgias,
  - Peripheral neuropathy
  - Glomerulonephritis.
  - Mixed cryoglobulinemias (positive RF or inflammatory markers (complement factors C3/C4, CRP, ESR) may be found

ACUTE HEPATITIS

Incubation phase	4 to 12 weeks but can be up to 6 months.
Symptomatic hepatitis	•Can last from 4–12 weeks. •Symptoms (*fever, fatigue, nausea, dark urine, jaundice, myalgia and right upper quadrant abdominal pain.* •Symptoms are common in adults but not in infants and children.
Recovery period	This results in the normalisation of ALT levels.
Clearance phase	The HBsAg clears from the serum after a few months. It coincides with the development of anti-HBs.

CHRONIC HEPATITIS

Natural history and assessment of patients with chronic HBV infection

HBV markers

- HBsAg
- HBeAg/anti-HBe
- HBV DNA

Liver disease

- Biochemical parameters: ALT
- Non-invasive markers of fibrosis (elastography or biomarkers) *or* liver biopsy in selected cases

CHRONIC HEPATITIS

	HBeAg positive		HBeAg negative	
	Chronic infection	**Chronic hepatitis**	**Chronic infection**	**Chronic hepatitis**
HBs Ag	High	High/intermediate	low	intermediate
Hbe Ag	Positive	Positive	Negative	Negative
HBV DNA	$>10^7$ IU/ml	10^4-10^7 IU/ml	< 2000 IU/ml	> 2000 IU/ml
ALT	Normal	Elevated	Normal	Elevated
Liver disease	None/minimal	Moderate/severe	None	Moderate/severe
Old terminology	Immune tolerant	Immune reactive HBeAg positive	Inactive carrier	HBeAg negative chronic hepatitis

HEPATOCELLULAR CARCINOMA

- Group 1 human carcinogen with sufficient evidence to prove its causation of HCC (up to 233 fold increased risk)

Who should be screened?

- Pregnant women
- HIV, HCV patients
- Needlestick victims
- People undergoing dialysis
- Sexual assault victims
- MSM
- People in or who have ever been in prison
- People born in intermediate and high risk regions
- patients with elevated ALT or AST
- Person who inject drug (PWID)
- HCPs who perform exposure prone procedures
- Persons needing immunosuppressive therapy
- Heterosexuals with multiple sex partners

SCREENING TEST

- ***Hepatitis B surface antigen (HBsAg):-*** can be detectable during weeks 4–10.
- ***Antibody to surface antigen (anti-HBs):-*** is a protective antibody that develops with the resolution of acute infection or following successful vaccination against HBV.
- ***Antibody to core antigen (anti-HBc):-*** Anti-HBc immunoglobulin G (IgG) remains positive for life following exposure to HBV; however, unlike anti-HBs, anti-HBc is not a protective antibody, and this test may occasionally give a false-positive result.



PREVENTION OF HBV

HBV vaccine

Hepatitis B immunoglobulins

HBV Vaccine

- It is inexpensive, safe and effective.
- The currently available HBV vaccines consist of recombinant HBsAg produced in yeast.
- A series of 3 intramuscular injections may achieve.
- Anti-HBs consider protective if levels >10 mIU/mL and can be reached in approximately 85-95% of vaccinated individuals.
- Booster dose is controversial
- Higher doses may be needed with patient on dialysis or will receive immunosuppression therapy

HBV Vaccine

- *Single-antigen hepatitis B vaccines*
  - *RECOMBIVAX* HB®
  - *Engrix-B®*
- *Combination vaccines*
  - *PEDIARIX*®: Combined hepatitis B, diphtheria, tetanus, acellular pertussis (DTaP), and inactivated poliovirus (IPV) vaccine. Cannot be administered before age 6 weeks or after age 7 years.
  - *TWINRIX*®: Combined Hepatitis A and hepatitis B vaccine.Recommended for people aged ≥18 years who are at increased risk for both HAV and HBV infections.

HEPATITIS B IMMUNOGLOBULIN

- Hepatitis B immunoglobulin (HBIG) is prepared from pooled plasma, with samples selected on the basis of high levels of anti-HBs.
- HBIG is recommended in:-
  - Infants born to HBsAg-positive mothers.
  - Non-immune people exposed to blood of chronic HBV people.
  - Post liver transplant to prevent re-infection



HEPATITIS B IMMUNOGLOBULIN

- For unvaccinated adults, HBIG should be given within 72 hours of exposure, because efficacy diminishes with time from 48 hours after exposure. The HBV vaccine can be given at the same time as HBIG or within 7 days of exposure.
- Hepabig 200 IU :- 0.06 mL/kg IM (only) once



HEPATITIS B IMMUNOGLOBULIN

Mechanism of Action

Pooled human specific immune globulins G from donors; nonpyrogenic solution that neutralize hepatitis B surface antigen

Pharmacokinetics

- Half-Life: 17-25 days
- Peak Plasma Time: 2-10 days (IM)
- Duration: 3-6 months (postexposure prophylaxis)

Management

Not all patients with hepatitis B need treatment

GOALS OF THERAPY

In acute hepatitis B, the main goal of therapy is to:-

I. Preventing the risk of acute or subacute liver failure.

II. Improving the quality of life by shortening the duration of the disease associated symptoms.

III. lowering the risk of chronicity.

GOALS OF THERAPY

In chronic hepatitis B, the main goal of therapy is to:-

I. Improve survival and quality of life by preventing disease progression, and consequently HCC development.

II. Prevent mother to child transmission.

III. Prevent hepatitis B reactivation.

IV. Prevention and treatment of HBV-associated extrahepatic manifestations.

Immune-modulator: IFN

Direct antiviral: Polymerase inhibitors (NA)

INTERFERON THERAPY

- Standard-IFN was the first therapy used for treatment of chronic HBV infection and has been used for over the last two decades.
- Mechanism of action: it binds to receptors on CM, initiates multiple complex mechanisms leads to:
    1. Suppression of viral protein synthesis.
    2. Degradation of viral mRNA.
    3. Prevention of viral infection of cells.
    4. Enhancement of antigen presentation by HLA I and II to the immune system.
    5. activation of natural killer and other immune cells, and increased cytokine production.

INTERFERON THERAPY

- The conjugation of the IFN-α molecule with polyethylene glycol, which decreases the renal excretion of IFN, thus prolonging its half-life and allowing for once weekly dosing (subcutaneously)) intervals.
  - Peak plasma time: 72-96 hr
  - Half-life: 160 hr
- Alfa interferons cause or aggravate fatal or life-threatening neuropsychiatric, autoimmune, ischemic, and infectious disorders

Assignment

Side effects of IFN

HBV POLYMERASE INHIBITORS: NUCLEOSIDE AND NUCLEOTIDE ANALOGUES

Mechanism of action

- Nucleos(t)ide analogues are selective competitive inhibitors of HBV polymerase. These agents are incorporated into the viral DNA strand, resulting in chain termination.
- Nucleos(t)ide analogues may interfere with the synthesis of the negative DNA strand by reverse transcription, synthesis of the positive DNA strand, and possibly cccDNA formation in newly infected cells.

HBV POLYMERASE INHIBITORS

- These agents are orally administered
- Infinite duration (compared to finite therapy with IFN-based regimen)

Nucleotide analogue (Nucleoside + 3Phosphatees linked)			
Tenofovir disproxil fumrate Tenofovir alphenamide Adefovir dipivoxil	TDF TAF ADV	300 mg 25 mg 10 mg	PO qDay
Nucleoside analogues:			
Lamivudine Entecavir	LAM ETV	100 mg 0.5 mg or 1 mg	PO qDay

HBV POLYMERASE INHIBITORS

Nucleotide analogue	Peak plasma level	Half Life	Excretion
Tenofovir disproxil fumrate	1h (fasting)	17 h	Urine (70-80%)
Tenofovir alphenamide	0.5 h	0.5 h	Feces (31%), urine (1%)
Adefovir dipivoxil	0.58-4 hr	8-9 h	Urine
Nucleoside analogues:			
Lamivudine	0.5-1.5 hr	2 h	Urine
Entecavir		5-6 day	Urine

HBV POLYMERASE INHIBITORS ADVERSE EVENT

Common Side Effects

- Fatigue
- Headache
- Nausea
- Diarrhea
- Abdominal pain
- Dizziness

HBV POLYMERASE INHIBITORS ADVERSE EVENT

Serious Side Effects

1.Nephrotoxicity (Renal Impairment)

1. Seen with **tenofovir (TDF)**
2. Can cause proximal renal tubulopathy and decreased bone mineral density
3. Monitor renal function (serum creatinine, eGFR, urine protein)

2.Lactic Acidosis and Hepatic Steatosis

1. Rare but serious (associated with **entecavir, lamivudine, telbivudine, adefovir**)
2. Symptoms: muscle pain, rapid breathing, fatigue
3. Discontinue if suspected

HBV POLYMERASE INHIBITORS ADVERSE EVENT

Serious Side Effects

3. **Myopathy and Neuropathy**

   Associated with **telbivudine**

4. **Bone Mineral Density Loss**

   Tenofovir (TDF) may reduce bone density (Consider T**enofovir alafenamide (TAF)** instead for lower bone toxicity

5. **Hepatic Flares After Discontinuation**

   Withdrawal of NAs can trigger severe hepatitis flares

6. **Pancreatitis** (rare)

   Reported with **lamivudine** in some cases

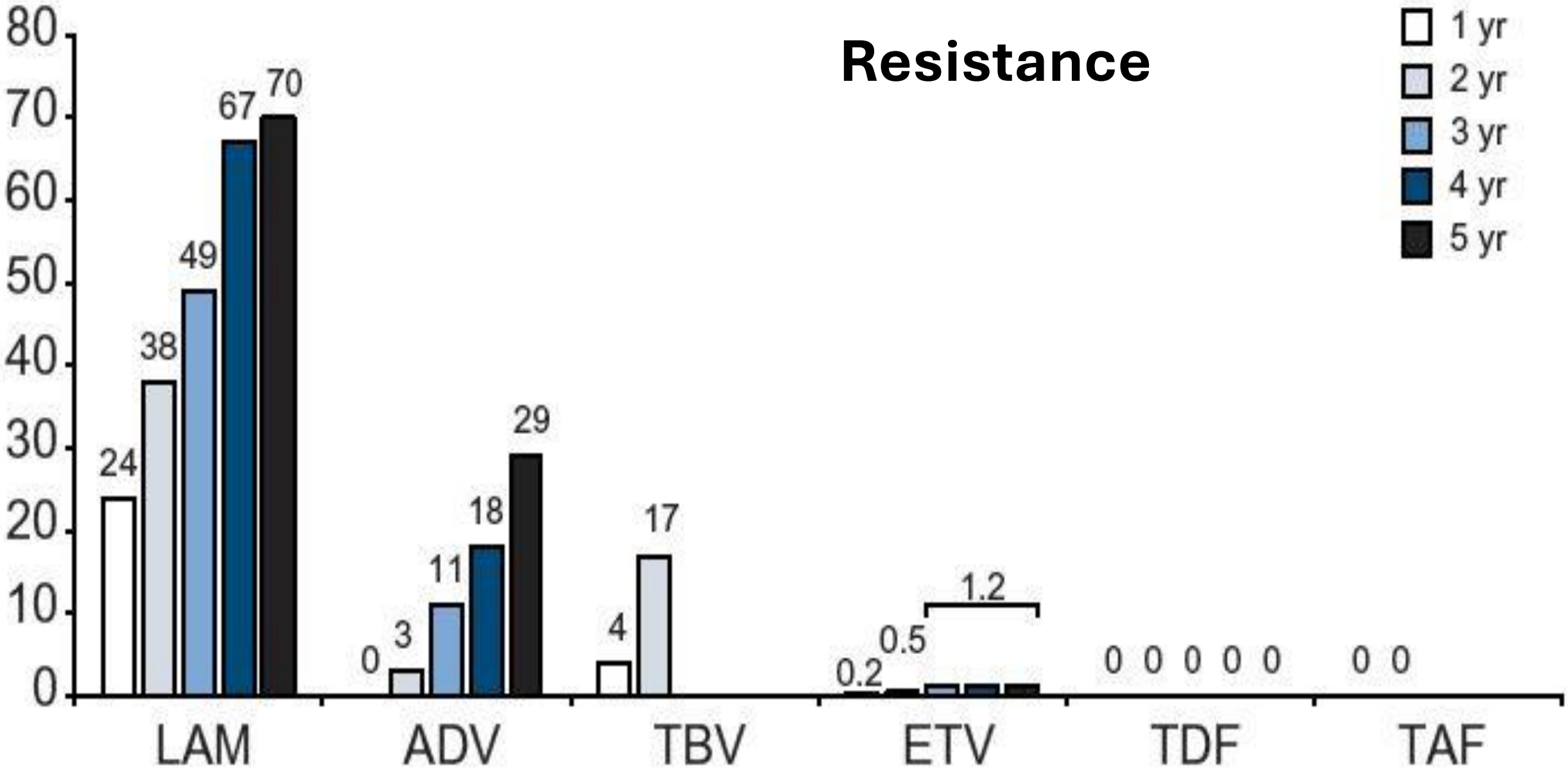
Patients with renal impairment, on dialysis:

eGFR (ml/min)	ETV	TDF	frequency
> 50	No dose adjustment		
30 -49	0.5 mg	300 mg	/ 2 days
10-29	0.5 mg	300 mg	/ 3 days
<10	0.5 mg	300 mg	Once weekly

In patient with LAM resistance or decompensated cirrhosis ETV dose should be doubled

IV. Patients with renal impairment, on dialysis:

- ETV (TAF) is preferred in naïve patient and should be adjusted according to eGFR
- Patient on TDF either to shift to TAF, or should continue with dose adjusted to eGFR (in severe cases may shift to entecavir)
- Patient undergoing dialysis, all agents should be given once weekly after the session


Resistance
1 yr
2 yr
3 yr
4 yr
5 yr
80
70
60
50
40
30
20
10
0
24
38
49
67
70
0
3
11
18
29
4
17
0.2
0.5
1.2
0 0 0 0 0
0 0
LAM
ADV
TBV
ETV
TDF
TAF

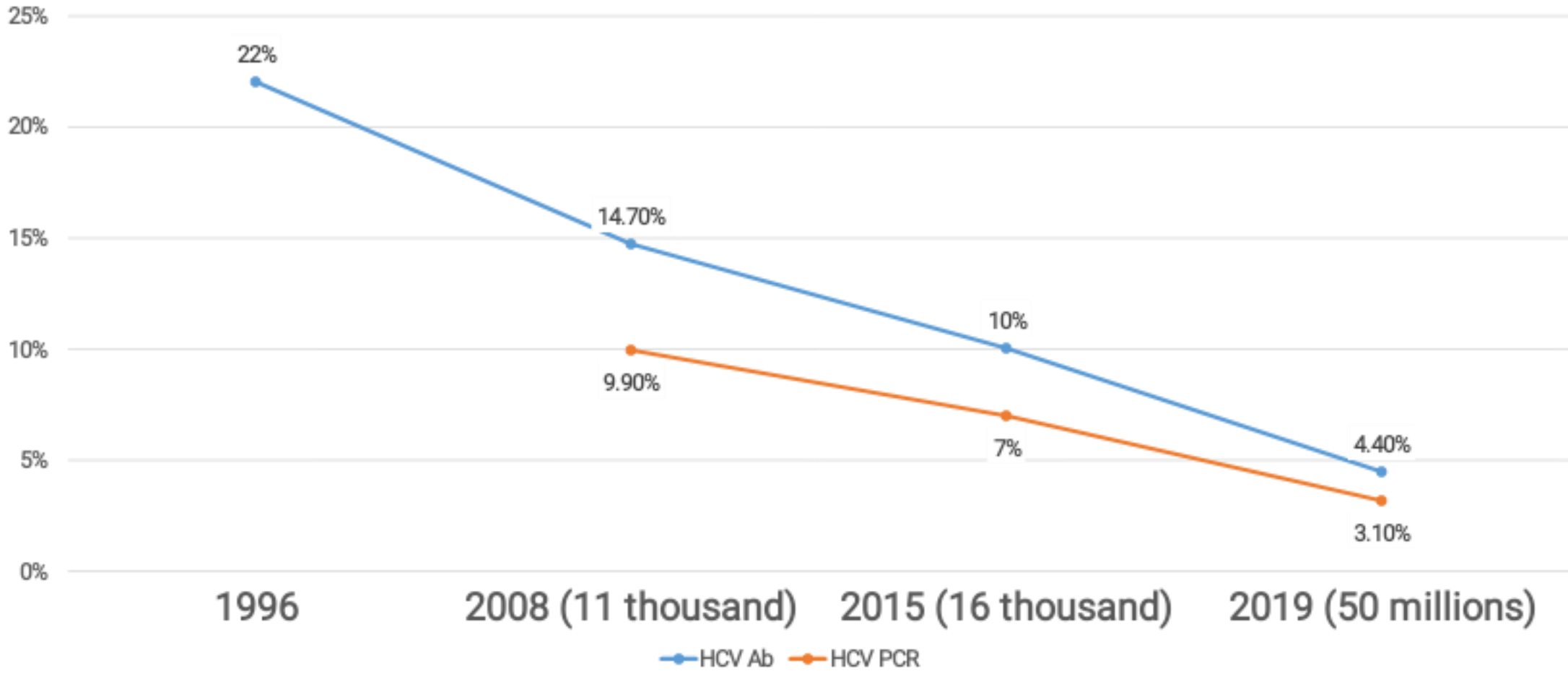
Chronic HCV

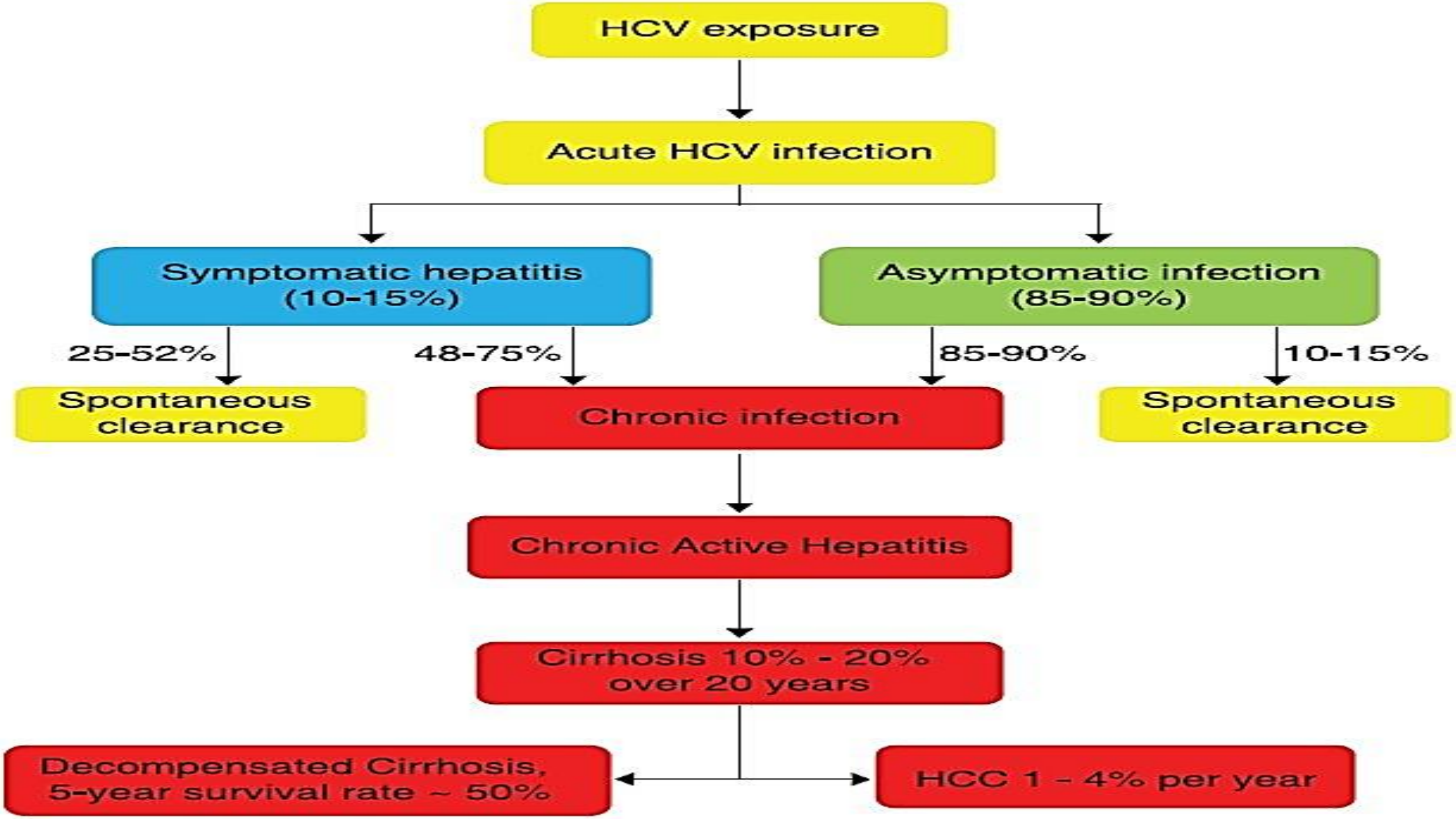
Background

- Egypt started getting worried about HCV in 1989.
- In 2007 National Committee for Control of Hepatitis was
- They conduct large multistage survey for HCV prevalence in 2008 & 2015
- Since October 2018, Egypt start the largest campaign in the world for HCV screening and treatment (plus non communicable disease)

"صحة مليون 100"

Estimated prevalence in adults in Egypt



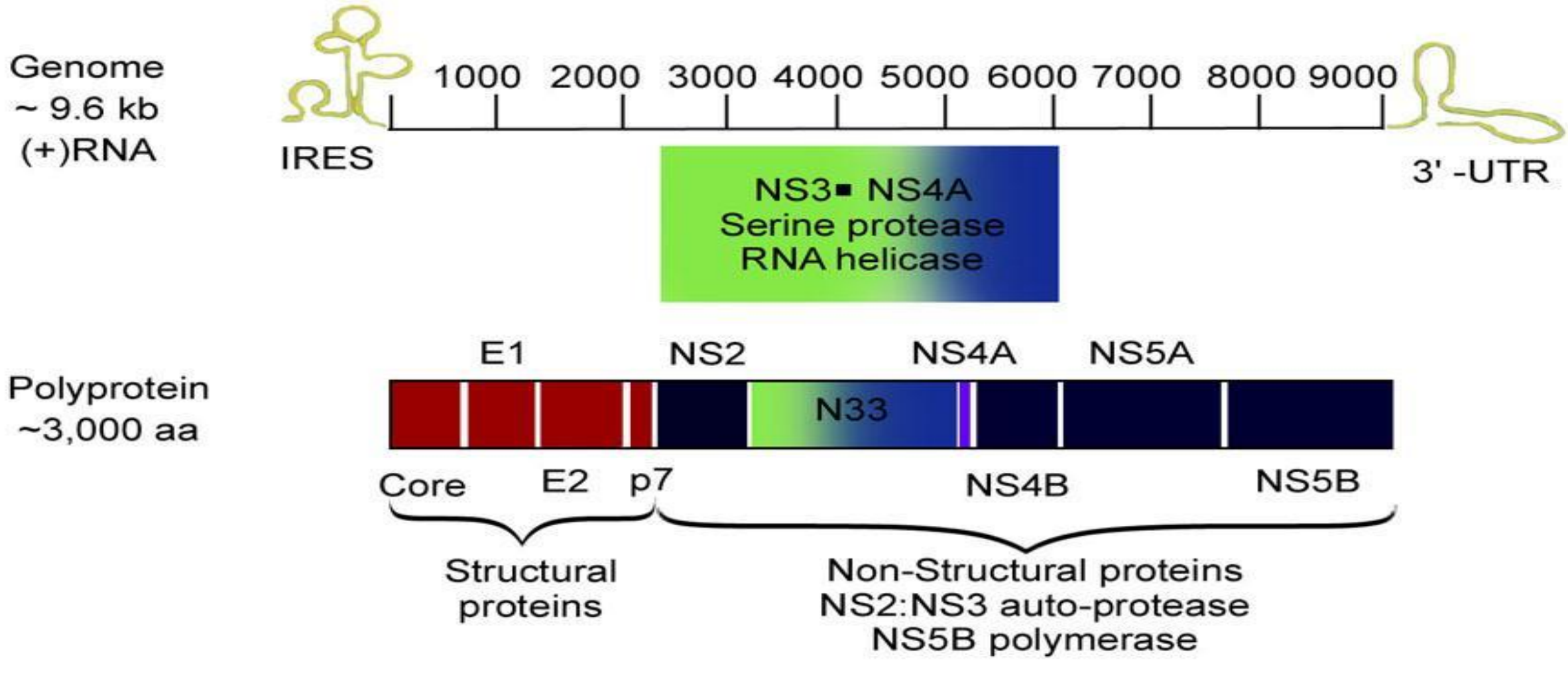

HCV exposure
Acute HCV infection
Symptomatic hepatitis (10-15%)
Asymptomatic infection (85-90%)
25-52%
48-75%
85-90%
10-15%
Spontaneous clearance
Chronic infection
Spontaneous clearance
Chronic Active Hepatitis
Cirrhosis 10% - 20% over 20 years
Decompensated Cirrhosis, 5-year survival rate ~ 50%
HCC 1 - 4% per year

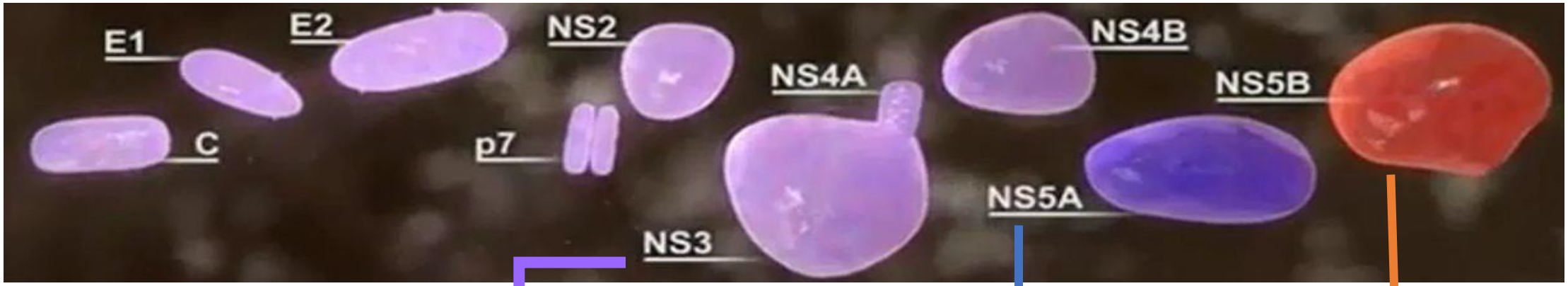
All patients with hepatitis C should be treated with some exception

Who should be excluded from treatment

- End stage liver disease (except if prepared for liver transplantation)
- Pregnancy, lactation or inability to use effective contraceptive
- Malignancy
  - HCC except 6 months after ablation
  - Extra-hepatic malignancy after 2 years of disease-free interval (in lymphoma and CLL treatment can be initiated immediately after remission

HCV Genome and Polyprotein Potential Drug Targets





NS3/4A (Protease) inhibitors

.....previr

- Telaprevir
- Boceprevir
- Simeprevir
- Asunaprevir
- Danoprevir
- **Paritaprevir**
- **Voxilaprevir**
- **Grazoprevir**
- **Glecaprevir**

NS5A inhibitors

.....asvir

- **Daclatasvir**
- **Ledipasvir**
- **Ombitasvir**
- **Velpatasvir**
- **Elbasvir**
- **Pibrentasvir**
- Samatasvir

NS5B (polymerase) inhibitors

....buvir

- **Nucleotide inhibitor**
  Sofosbuvir
- **Non nucleotide inhibitor**
  Dasabuvir

Active ingredients	Brand (trade) name	Drug concentration
Sofosbuvir *	→ Sovaldi	400 mg
Daclatasvir *	→Daklinza	60 mg
Ledipasvir/Sofosbuvir *	→Harvoni	90/400 mg
Paritaprevir/ritonavir/Ombitasvir#	→Qurevo	75/50/12.5 mg
Sofosbuvir/Velpatasvir*	→Epclusa	400/100 mg
Sofosbuvir/Velpatasvir/Voxilaprevir*	→Vosevi	400/100/100 mg
Elbasvir/Grazoprevir¶	→Zepatir	50/100 mg
Glecaprevir/Pibrentasvir¶	→Mavyret	100/40 mg
Ribavirin*	→Copegus	200 or 400 mg

* Available in Egypt
Not used now
¶ Available Ineurope and USA

Sofosbuvir (Sovaldi)

Mechanism of Action

- Nucleotide prodrug that undergoes metabolism to the active uridine analog triphosphate, an inhibitor of HCV NS5B RNA-dependent polymerase; its inhibition in turn suppresses viral replication



Pharmacokinetics

Absorption

Peak plasma time: 0.5-2 hr (sofosbuvir); 2-4 hr (metabolite GS-331007)

Distribution

Plasma bound: 61-65% (sofosbuvir); minimal for metabolite GS-331007

Pharmacokinetics

Metabolism

Liver

Elimination

- Excretion: Urine (78% metabolite GS-331007; 3.5% sofosbuvir)
- Half-life: 0.4hr (sofosbuvir); 27 hr (metabolite GS-331007)

Dosage Modification

- Reduction of sofosbuvir dose not recommended
- Discontinue sofosbuvir therapy if the agents used in combination are discontinued (not be used as monotherapy)

Interactions

Amiodarone

(black box warning)

- sofosbuvir increases toxicity of amiodarone by unknown mechanism. Avoid or Use Alternate Drug. Postmarketing reports describe bradycardia resulting in death or pacemaker insertion when amiodarone was coadministered with sofosbuvir in combination with another direct acting antiviral.

Interactions

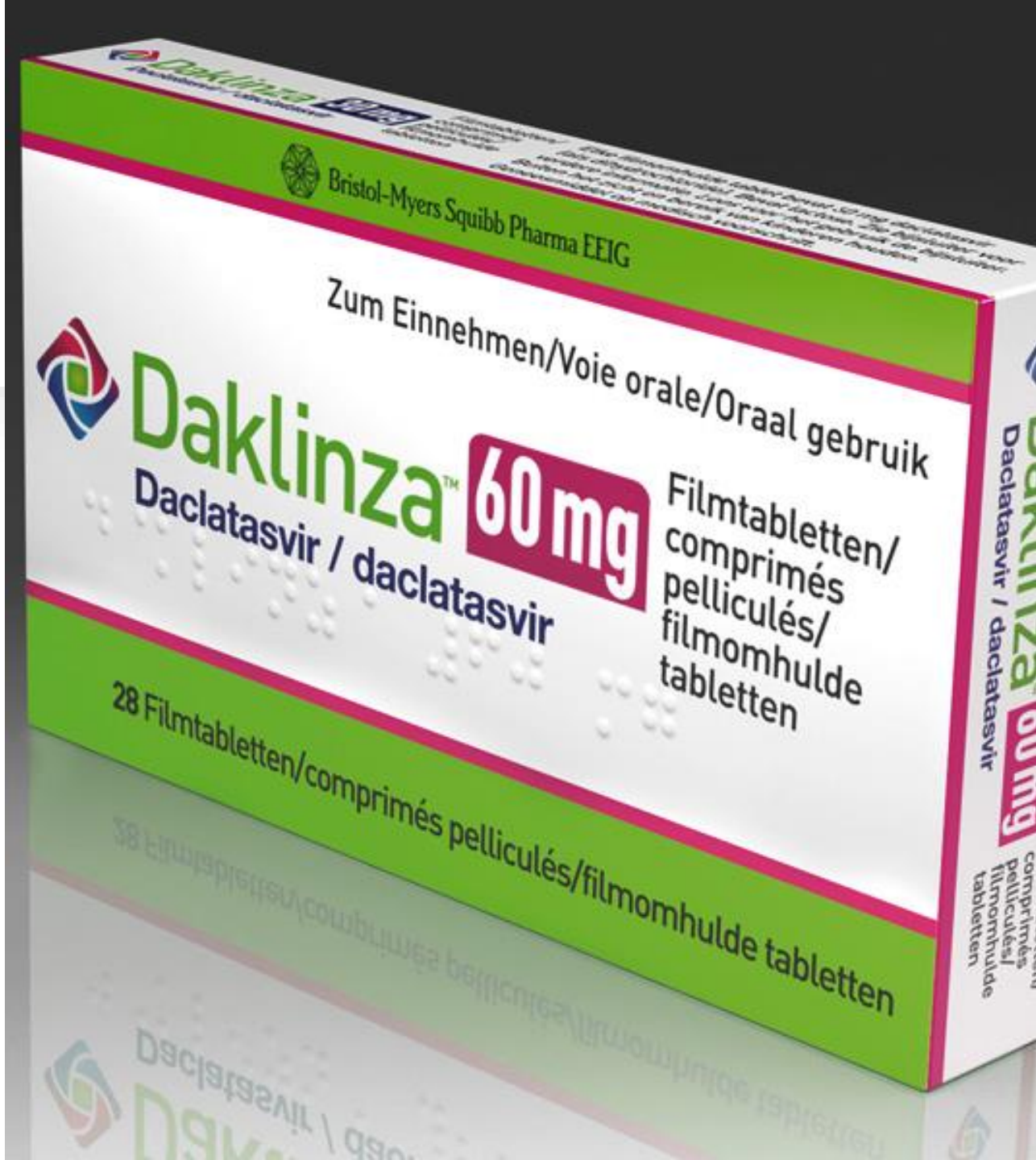
- carbamazepine
- oxcarbazepine
- phenobarbital
- phenytoin
- Rifampin

→ will decrease the level or effect of sofosbuvir by P-glycoprotein (MDR1) efflux transporter. Avoid or Use Alternate Drug.

HCV NS5A Inhibitors Daclatasvir (Daklinza)

Mechanism of Action

- Inhibits NS5A, a nonstructural protein encoded by HCV
- Binds to the N-terminus within domain 1 of NS5A, which may cause structural distortions that interfere with NS5A functions, and thereby inhibits both viral RNA replication and virion assembly



Pharmacokinetics

Absorption

- Absolute bioavailability: 67%
- Peak plasma time: 2 hr
- Peak plasma concentration: 182 ng/mL
- May take with or without food

Distribution

Protein bound: 99%

Pharmacokinetics

Metabolism

Substrate of CYP3A, with CYP3A4 being the primary CYP isoform responsible for metabolism

Elimination

- Half-life: 12-15 hr
- Excretion: 88% feces (53% of the dose as unchanged drug); 6.6% urine (primarily unchanged drug)

ledipasvir/sofosbuvir (Harvoni)

Mechanism of Action

- **Ledipasvir**: Inhibits HCV NS5A protein, which is required for viral replication; resistance selection in cell culture and cross-resistance studies indicate ledipasvir targets NS5A as its mode of action
- **Sofosbuvir**: described previously



Pharmacokinetics

Absorption

Peak plasma concentration

- ledipasvir: 4-4.5 hr
- sofosbuvir: 0.8-1 hr
- GS-331007: 3.5-4 hr

Peak plasma level

- ledipasvir: 323 ng/mL
- sofosbuvir: 618 ng/mL
- GS-331007: 707 ng/mL

Pharmacokinetics

Distribution

Protein bound

- ledipasvir: >99.8%
- sofosbuvir: 61-65%
- GS-331007: minimal

Metabolism

- ledipasvir: No detectable metabolism
- sofosbuvir: Extensively metabolized in liver to form the pharmacologically active nucleoside analog triphosphate GS-461203; dephosphorylation results in formation of GS331007 (accounts for <90% of total systemic exposure of sofosbuvir)

Pharmacokinetics

Elimination

Half-life, terminal

- ledipasvir: 47 hr
- sofosbuvir: 0.5 hr
- GS-331007: 27 hr

Excretion

- ledipasvir: >98% feces (mostly unchanged); <1% urine
- GS-331007: 14% feces; 80% urine; 2.5% expired air

Sofosbuvir/Velpatasvir (Epclusa)

Mechanism of Action

- **Velpatasvir**: Inhibits HCV NS5A protein, which is required for viral replication; resistance selection in cell culture and cross-resistance studies indicate ledipasvir targets NS5A as its mode of action
- **Sofosbuvir**: described previously



Pharmacokinetics

Absorption

Velpatasvir solubility decreases as gastric pH increases; practically insoluble (<0.1 mg/mL) at pH >5 (avoid PPI)

Peak plasma concentration

- Velpatasvir: 3 hr
- sofosbuvir: 0.8-1 hr
- GS-331007: 3.5-4 hr

Peak plasma level

- Velpatasvir: 259 ng/mL
- Sofosbuvir: 567 ng/m
- GS-331007: 707 ng/mL

Pharmacokinetics

Distribution

Protein bound

- Velpatasvir: >99.5%
- sofosbuvir: 61-65%
- GS-331007: minimal

Metabolism

- Velpatasvir: Substrate of P-gp transporter, CYP2B6, 2C8, and 3A4
- sofosbuvir: Extensively metabolized in liver to form the pharmacologically active nucleoside analog triphosphate GS-461203; dephosphorylation results in formation of GS331007 (accounts for <90% of total systemic exposure of sofosbuvir)

Pharmacokinetics

Elimination

Half-life, terminal

- Velpatasvir: 15 hr
- Sofosbuvir: 0.5 hr
- GS-331007: 27 hr

Excretion

- ledipasvir: 94% feces (mostly unchanged); 14% urine
- GS-331007: 14% feces; 80% urine; 2.5% expired air

HCV genotype 4 infection TTT
Conclusion

Several options with similar efficacy are recommended for treatment-naive patients with HCV genotype 4 infection, as follows:

Daily fixed-dose combination of

- Sofosbuvir (400 mg)/ Daclatasvir (60 mg).
- Ledipasvir (90 mg)/sofosbuvir (400 mg)
- Sofosbuvir (400 mg)/ Velpatasvir (100mg).

All used for 12 weeks in naïve patient without liver cirrhosis or with compensated liver cirrhosis

- In decompensated cirrhosis weight-based ribavirin (RBV) (1000 mg [<75 kg] to 1200 mg [≥75 kg]) is added or extended treatment for 24 weeks if RBV is contraindicated.

DAAs Adverse Events

Common Side Effects

- Fatigue
- Headache
- Nausea
- Diarrhea
- Insomnia

Black Box Warnings *(For all drugs)*

- Test all patients for evidence of current or prior hepatitis B virus (HBV) infection before initiating treatment with HCV direct acting antivirals (DDAs)
- HBV reactivation has been reported in HCV/HBV coinfected patients who were undergoing or had completed treatment with DDAs and were not receiving HBV antiviral therapy

- Some cases have resulted in fulminant hepatitis, hepatic failure, and death
- Monitor HCV/HBV coinfected patients for hepatitis flare or HBV reactivation during HCV treatment and post-treatment follow-up
- Initiate appropriate patient management for HBV infection as clinically indicated

Assignment

- Dermatological side effect (especially with protease inhibitors)
- Other drug-drug interaction (Statins)



Thank you

